

Shush! Don't tell any windows... domain

Level 3: Step 2

HARD

Theme: Networked Systems: NFS & SMB

Due Date: 20/01/2023

Please take the following notes in consideration:

- In this assignment the questions marked with a (*) are practical and have to be shown to the teacher (no report is needed).
- The remaining questions are to be answered in a written report, to be delivered to the teacher.
- Whenever possible, the VMs necessary to resolve the practical questions should be created as *linked clones*.

Authentication Domains

1. Investigate the role of **Active Directory** in the **Windows Server** operating systems and how they can leverage User and Computer Management in networks with a large number of computers.
2. What is a Domain Controller? Briefly describe how a workstation joining a given domain, finds the Domain Controller.
3. Can you create a Domain Controller using the Linux Operating System? Which implementation of the **SMB** protocol for linux would allow you to do this?
4. (*) Create a **Debian Linux v11** VM.
 - (a) Rename the computer to **gsr-dc**
 - (b) Determine the subnet, gateway and dns server settings provided by your hypervisor to the newly installed VM and set a static IP address for it (ex.: 192.168.122.150)
 - (c) Install **Samba 4** (`samba` package) and **Windbind** (`winbind` package).
 - (d) Disable and stop the *standalone smbd* service and unmask/enable the **samba-ad-dc** service
 - (e) Provision **Samba 4** as an **Active Directory Domain Controller** with the parameters:
 - **Realm:** gsr.ipb.local
 - **Domain:** gsr
 - **Server Role:** dc
 - **Dns Backend:** SAMBA_INTERNAL
 - **DNS forwarder IP:** the dns server you retrieved in (b)

- (f) Start the **samba-ad-dc** service
- 5. (*) Create (or clone) a VM with **Windows 10** and perform the following operations:
 - (a) Rename the computer to **gsr-ws-01**
 - (b) Make sure that **gsr-ws-01** is using **gsr-dc** (from the previous question) as the primary DNS Server
 - (c) Make the workstation join the **gsr.ipb.local** domain and login as the **GSR\Administrator** user
 - (d) Install the official **Remote Server Administration Tools** and open **Active Directory Users and Computers**
 - Verify that you have management access to the **gsr.ipb.local** domain
 - (e) Under **gsr.ipb.local\Users** create:
 - A domain group named **GSR Users**
 - A user for each student in the working group (ex.: a12345, m300321)
 - Add the 2 users to the **GSR Users** group
 - Make sure all the newly created users are part of the **Domain Users** group
 - (f) Demonstrate that all the newly created domain users can login to this computer

Network File System

- 6. When mounting **NFS** shares, the local user and group ids in the client and server may differ. What NFS mechanism allows you to map ids between the two hosts?
- 7. (*) Create (or clone) a VM with **Debian Linux v11** and perform the following operations:
 - (a) Set a static IP address for the computer (ex: 192.168.122.180)
 - (b) Create the `/home/gsr` directory
 - (c) Install the `nfs-kernel-server` package and make the necessary changes to the `/etc/exports` file so that the `/home/gsr` directory is shared **rw** (read/write) the **gsr-dc** server.
- 8. (*) Back in the `gsr-dc` server, perform the operations:
 - (a) Make the necessary changes to `/etc/fstab` to mount the share created in 7.c) into `/mnt/gsr` **at boot time**
 - (b) Share this folder as `\\gsr-dc\gsr` (read only)
 - (c) Demonstrate the folder is accessible to all domain users.

◇ Keywords ◇

- samba 4 active directory domain controller
- active directory domain services
- remote windows administration tools
- server manager add feature
- nfs kernel server debian
- nfs id mapping
- fstab syntax nfs

Deliverables

9. All deliverables submitted via [discord#assignments-submission](#)):

1. PDF file with the report
2. The file name should be 13s2h_aXXXXX_aYYYYY.pdf. This code means:
 - Level 3
 - Step 2
 - Easy difficulty
 - Students that participated in the group
 - **Maximum:** 2 students